

Sayed Mohammed Asim

Full Stack Software Engineer & AI Enthusiast (or Software Engineer & AI Developer)

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PROFESSIONAL SUMMARY

Proactive 3rd-year Computer Science & Engineering student at Chittagong University of Engineering and Technology (CUET) with practical expertise in Full Stack Web Development, Artificial Intelligence & Computer Vision, and High-Performance Backend Architectures. Proven track record of developing gesture-based driving controllers, OCR-driven sentiment emotion recommendation engines, hackathon REST APIs, and scalable Node.js backend systems. Seeking Software Engineer & AI Developer internship and full-time opportunities.

TECHNICAL SKILLS

Programming Languages
Python, JavaScript (ES6+), Dart, Java, C/C++, HTML5/CSS3

Databases & Tools
PostgreSQL, SQLite, Docker, Git/GitHub, Jest Testing, Linux/Shell

AI & Computer Vision
OpenCV, MediaPipe, Tesseract OCR, Sentiment Analysis / NLP, Scikit-Learn

Backend & Web Dev
Node.js, Express.js, RESTful APIs, React.js, Redis, Modular Architecture

Mobile Development
Flutter, Firebase Realtime DB, Geolocation Services

FEATURED PROJECTS

VirtualWeel — Gesture Controlled Driving System
Tech Stack: Python, OpenCV, MediaPipe, PyAutoGUI GitHub: https://github.com/smasim825/VirtualWeel
- Engineered a real-time computer vision application that translates hand gestures into driving control inputs (W, A, S, D) for video games.
- Utilized MediaPipe for low-latency 2D landmark extraction and OpenCV to render a dynamic HUD steering wheel overlay.

Music Mood Recommender — OCR & Sentiment AI Engine
Tech Stack: Python, Tesseract OCR, NLP Sentiment Analysis, Spotify API, Flask GitHub: https://github.com/smasim825/mood-music-recommender
- Built an intelligent context-aware music recommendation system that extracts text excerpts from images/book pages via Tesseract OCR.
- Classifies underlying textual emotion (happy, sad, romantic, calm) and queries Spotify API for matching playlists.

Sentinel Safety — Mobile Safety Platform
Tech Stack: Flutter, Dart, Firebase, Geolocation APIs GitHub: https://github.com/smasim825/sentinel-ai-1.0
- Developed a cross-platform mobile safety application providing real-time GPS tracking and emergency assistance.
- Implemented automated distress alerts, guardian notifications, and cloud database synchronization via Firebase.

CoWork — Multi-Tenant Coworking Space Booking API
Tech Stack: Python, REST API, PostgreSQL, Docker, JWT GitHub: https://github.com/smasim825/ICT_Fest_Hackathon_Preliminary
- Developed for ICT Fest Hackathon Preliminary: A multi-tenant REST API managing coworking space allocations across multiple organizations.
- Features role-based access control (Admin vs Member), time-slot overlap prevention, and occupancy analytics.

Master Backend Architecture & Revision System
Tech Stack: Node.js, Express.js, Redis, Jest, Security/JWT GitHub: https://github.com/smasim825/Backend_Development
- Built an enterprise-grade backend architecture reference implementation demonstrating separation of concerns (app.js vs server.js).
- Implemented Redis caching layers, middleware validation, security hardening, and Jest automated test suites.

EDUCATION & ACHIEVEMENTS

B.Sc. in Computer Science & Engineering (3rd Year Undergraduate), *Chittagong University of Engineering and Technology (CUET)* 2027
Expected Graduation: 2027 Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Vision, Software Architecture.
- ICT Fest Hackathon Preliminary Contender.
- Competitive Programming & Problem Solving: Maintained cp_shortcuts repository covering graph algorithms, dynamic programming, and data structure templates.